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Data-based Analysis of Power Generation and Transmission Losses in Nigeria

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***Abstract*—The availability and reliability of power supply are vital to socioeconomic development. Nigeria is the most populous country in Africa with electrical power generation that is just a fragment of the energy demands of the populace. In this study, we analyzed the trend of power generation and losses in the Nigerian power system. The average peak power generated over the study period is just 4229.39MW which is approximately 23.5MW per million people. This is meagre when compared with about 500MW per million people in South Africa. Therefore, adequate and sustainable effort are required in terms of investment and management at the generation, transmission and distribution levels toward ensuring improved power supply in Nigeria. There is a need to introduce renewable energy sources into the energy mix at the national level in order to reduce environmental pollution, and also to promote the use of sustainable alternative energy sources at the household and industrial level which will stimulate private sector participation through the establishment of commercial energy farms.**

***Index Terms*--Energy audit, power distribution network, power system reliability, Nigeria's electricity generation, supply reliability**

I. INTRODUCTION

The population of Nigeria is constantly on the increase, and it is currently estimated that there are 180 million Nigerians. Increasing population size implies increasing energy demand [1]. Power generation and availability varies from country to country, and for an industrialized city in a developed country like New Zealand, Saudi Arabia and United States, more than 1,000MW is available per one million people while in Malaysia and South Africa about 500MW is generated per one million people. Sadly, for a number of African countries like Guinea-Bissau, Liberia, Sierra Leone, Niger, Nigeria etc. power generation is less than 30MW per million people resulting in various levels of energy poverty [2].

The Nigerian power sector is bedevilled by a number of challenges: economic, technical, managerial, corruption [1], policy [3, 4] etc. which has resulted in several decades of decay and poor power infrastructure. The dwindling quality and quantity of electrical energy generated and distributed in Nigeria has greatly inhibited national growth, and socio-economic development [5, 6]. Power availability and reliability are vital to economic development and technological advancement. This reality has inhibited the socio-economic and technological development of Nigeria.

In 2013, the power sector under the Power Holding Company of Nigeria was privatized by unbundling into 6 generation, 1 transmission and 11 distribution companies which were placed under private management, and the Nigerian Electricity Regulatory Commission (NERC) was given the regulatory oversight over the sector. NERC was established in March 2005 by the Electric Power Sector Reform Act 2005. Since the privatisation, the expectation of the Nigerian people in terms of power availability and reliability has been met with gross disappointment. Power generation has not significantly improved and there are major issues concerning the availability and installation of prepaid meters for all customers has mandated by NERC .

In 2015, Nigeria's power generation capacity stood at 12,500MW of which only about 7,140MW capacity was available, out of which the total operational capacity was 3,900MW [5, 7]. In 2019, with multiple efforts of the government to increase power generation through the commissioning of small-scale power plants at different parts of the country, the impact on daily energy availability in most households is yet to be felt. This is further aggravated by inadequate demand for electricity by the Power Distribution Companies (DisCos) resulting in constrained power generation below the available capacity. The low demand for energy by DisCos is as a result of the challenge faced during revenue collection with a number of customers not willing to pay due to the claim of excess charge for energy that was never available in the first place. Coupled with this, is the fact that a number of households are unmetered and different forms of energy theft exist within the distribution network. As observed by the Minister of Power, Works and Housing, the problem is gradually shifting from "no power" to "more power" that cannot be sold due to low demand. Figure 1

presents the map of the DisCos in Nigeria across the geopolitical zones.

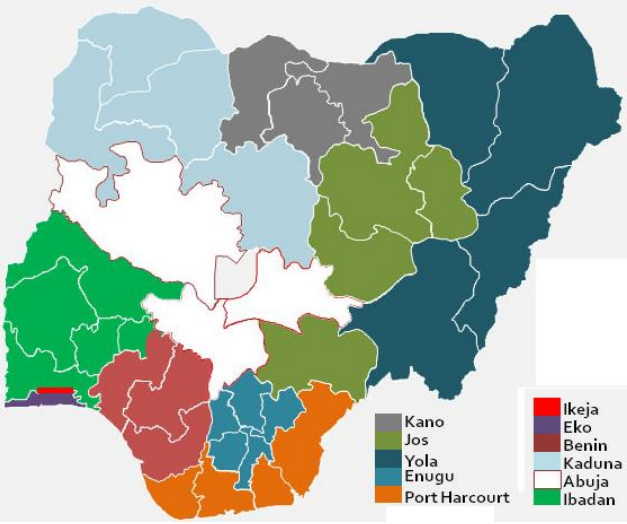


Figure 1. Map of the coverage area of each DisCo within Nigeria [8]

To further improve generation and distribution efficiencies, it is vital to start deploying off-grid renewable energy solutions [9] for selected areas with adequate revenue management policies put in place. This will also help to reduce the emission levels associated with non-renewable energy sources [10]. The economics of energy ought to be extensively studied considering the peculiarities of the Nigerian energy market in terms of poor metering networks, energy theft, unreliable supply, inadequate infrastructure, customer behaviour and frustration, lack of capital for needed development etc. The knowledge acquired from such detailed studies will further help in setting up adequate regulatory structure, tax, subsidy and pricing mechanism for a sustainable development [11]. To develop an effective energy management structure, it is important to consider electricity management, both from the demand aspect (customer usage and behaviour) and supply aspect (generation, transmission and distribution) [12, 13].

In this study, we considered the power generation trend in Nigeria by analysing the power generation data provided by NERC [14] from August 2016 to august 2018. The variation of the peak generated power across the period reveals interesting trends which is an indication of the low level of power reliability within the country. Also, the Aggregate Technical, Commercial and Collection (ATC&C) losses which measure losses in terms of revenue and energy losses show that the associated losses in the power sector are currently too high for sustainable development and productivity in the power sector. A number of customers are currently not metered and are therefore placed on estimated billing while others are even yet to be captured on the customer database and as such, energy is completely free for such users. The meters in some households are old and inaccurate giving readings that are far below the consumption level. Some customers are wrongly classified while others never receive bills due to bills distribution issues. For fairness, and to ensure the willingness of customers to pay, an appropriate billing system must be put in place [15].

II. STATISCAL ANALYSIS OF THE POWER GENERATION DATASET

The daily peak power generated (MW) is an indication of the operational capacity of the Nigerian power plants. This section presents the daily peak power trends over a period of 2 years from August 2016 to August 2018 as shown in Figure 3. Table I presents the descriptive statistics of the peak power generated daily. Figure 3 reveals a significant variation in peak power generated across the 2-year period which indicates power reliability issues that may be due to several instances of plant shutdown, disruption to gas supply, low level of water for hydro plants etc. The peak power generated within the period is 5,222MW.

Figure 2 presents a plot showing days when the transmission line voltage fluctuation was beyond the statutory 330kV (+/- 16.5) [14], and days with zero voltage indicating grid collapse are also displayed. Figure 4 shows the variation of the peak power across the four quarters in a year. In Figure 4, it was observed that a very significant variation occurred in Q1 2017 as compared with the other quarters. In Q4 2017, the outlier indicated a sudden drop in peak power to 2443MW after a steady variation between 4000MW and 5000MW. This is most likely due to plant shutdown or a systemic fault.

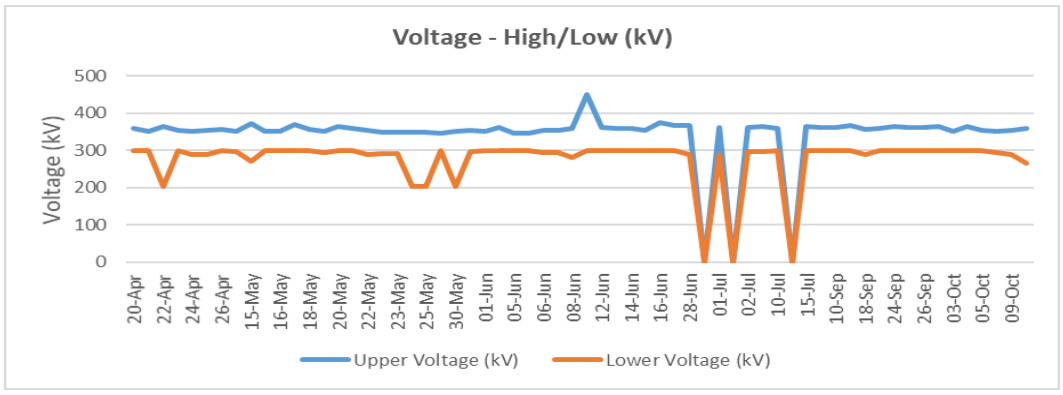


Figure 2. Transmission line voltage fluctuations [14]

TABLE I. DESCRIPTIVE STATISTICS OF DAILY PEAK POWER GENERATED

Min	Max	Range	Median	Mode	Mean	Std. error of mean	Standard Deviation	Variance
2,434 MW	5,222 MW	2,788 MW	4,273 MW	4034 MW	4229.39 MW	17.026 MW	469.7043 MW	22,0622.1 MW

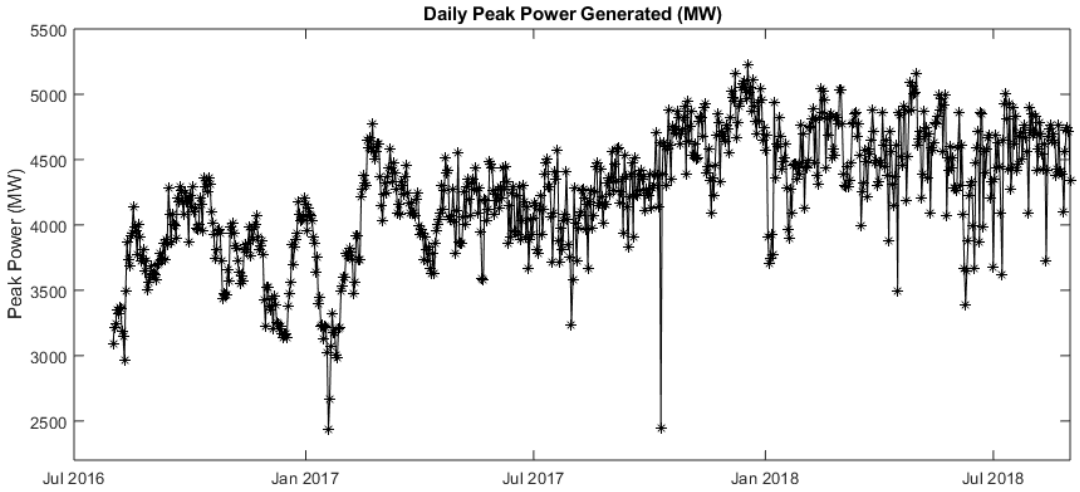


Figure 3. Daily peak power variation from 2016 to 2018

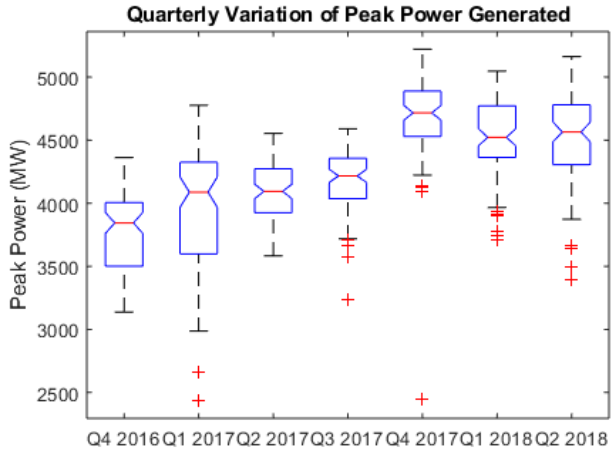


Figure 4. A box plot showing the variation of the daily peak power across yearly quarters from 2016 to 2018

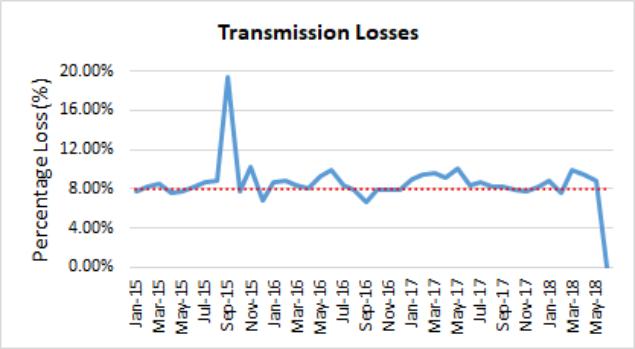


Figure 5. Transmission losses

III. ENERGY LOSSES IN THE NIGERIAN POWER SYSTEM

Increasing power generation without adequate measures to manage and minimise inherent system losses may result in futility and wastage. The percentage of the transmission losses in the Nigerian power system from 2015 to 2018 is shown in Figure 5. The chart shows variations in transmission losses below and above the Multi-Year Tariff Order (MYTO) target of 8.05%.

The Aggregate Technical, Commercial and Collection (ATC&C) losses is an index which measures power system losses in terms of energy and revenue losses. In Figure 6, the ATC&C losses incurred by each of the eleven DisCos is presented as a line chart from May 2015 to May 2018. The technical losses are due to losses in the transformers, equipment and transmission lines. The commercial losses include: energy theft, unmetered customers, metering errors etc. while uncollected revenue due to collection inefficiency accounts for the collection losses. The combination of these losses makes up the ATC&C losses which must be reduced to the barest minimum in order to maximize the returns on investment. At the kick-off of the privatization exercise on the 1st of November 2013, the DisCos were charged by their performance agreement to reduce their ATC&C losses to less than 20% within five years [16] but as shown by the line plot of Figure 6 and the box plot of Figure 7, the current trend reveals that ATC&C for Jos as at April 2017 was 81% and this was reduced to 79% in April 2018. A similar trend was observed for Kaduna with a peak ATC&C loss of 79% in October 2017. Eko and Ikeja DisCos in the western part of the country have a better performance when compared with the rest of the DisCos. The highest ATC&C loses incurred in Eko Disco was 42% in September 2017 and by April 2018 it had reduced to 30% and likewise, the highest ATC&C losses recorded in Ikeja was 58% with an average of 40.32% losses.

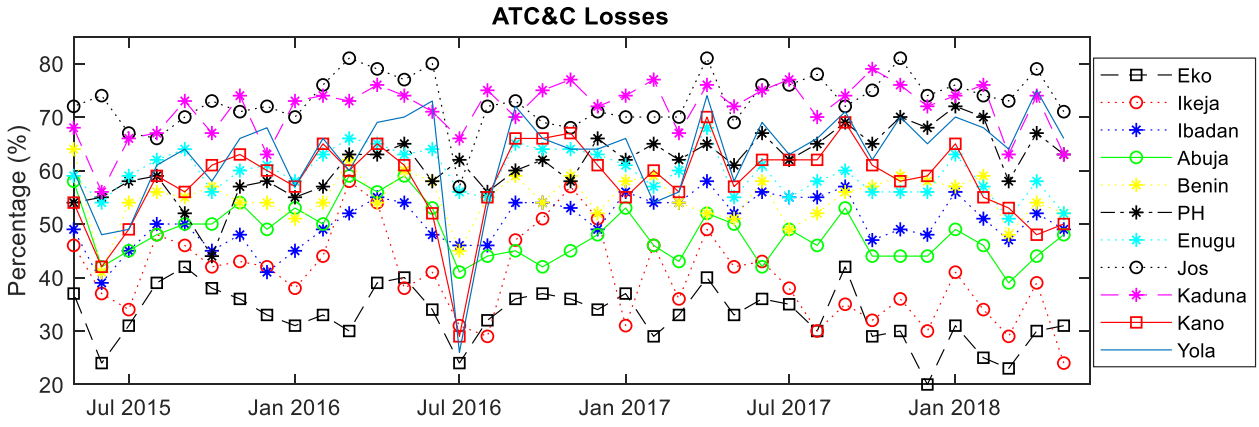


Figure 6. ATC&C losses for the eleven DisCos in Nigeria

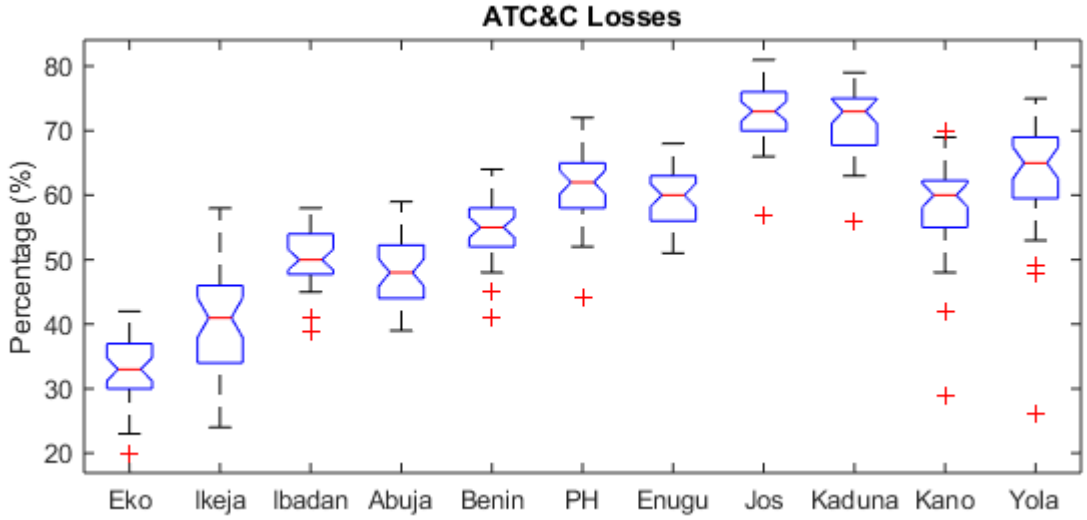


Figure 7. A box plot of the ATC&C losses for the eleven DisCos

Although Eko and Ikeja have relatively lower ATC&C losses but the current trend is still far from the 20% target. The contribution of each Disco to the total ATC&C losses over the period of study is presented as a pie chart in Figure 8. Eko DisCO accounts for 5% while Ikeja DisCo accounts for 7% of the total losses.

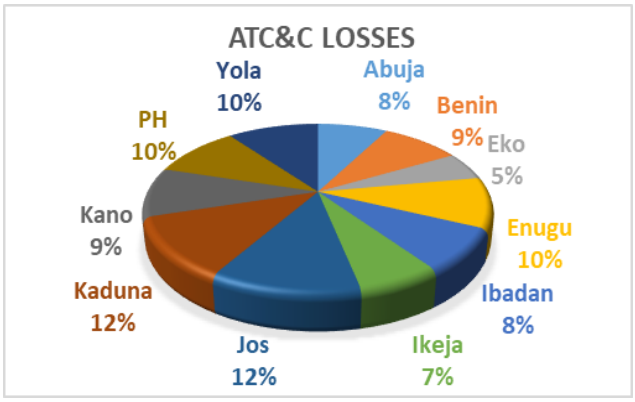


Figure 8. Percentage contribution of each Disco to the total ATC&C losses

IV. DISCUSSION

Despite the effort of successive governments toward improving power supply availability and reliability in Nigeria through privation, periodic injection of funds for infrastructural development, provision of regulatory oversight etc. Data trend reveals that the power sector has not significantly improved in terms of power generation, and technical, commercial and collection losses. As at 2015, the total operational capacity was 3,900MW while on the 31st of August 2018, the Daily Total Constrained Power was 4186MW, and this is only a slight improvement over the previous years. Also, in terms of the Aggregate Technical, Commercial and Collection Losses the current trend reveals that losses are still far above the 20% target. The implication of this operational reality is that the bulk of the energy is wasted in the form of energy losses on the network, uncollected payment for energy invoices, unmetered customers, inaccurate meters, energy theft etc. This is not a sustainable model because efforts and resources invested into the sector have not yielded meaningful results.

The Ministry of power is of the opinion that the DisCos have not demonstrated their commitment to the agreement at the time of privatization in terms of infrastructural development of the power system and provision of prepaid meters to all customers. The DisCos argue that they are short of funds needed for new infrastructural investments and are still currently servicing loans incurred when the privatization exercise commenced. As a result of the disagreement between the Federal Government represented by the Ministry of Power, Works and Housing and the investors in the DisCos; on the 31st of July 2018, the investors threatened to sell out the DisCos back to the Federal Government at a discounted price of 72 million US Dollar which is 10 million USD short of the price they bought it in 2013. This offer was based on their frustration as a result of the challenges in the industry and the lack of returns on investment. The investors claimed they were unable to carry out due diligence assessment of the physical asset at the point of purchase due to the protest by labour unions, and as a result, the investors were not adequately aware of the true status of things. In the interest of the nation, all concerned stakeholders need to conduct a detailed status assessment of the power sector to identify major challenges and sincerely commit to implement identified solutions. The strategy of shifting blames will not bring the desired change.

V. Recommendations

Improving the power availability and general operational efficiency of the power sector in Nigeria requires mitigating three key challenges, and these are: poor maintenance culture, inadequate finance and investment, and lack of a comprehensive power development master plan. Facility maintenance is generally a challenge in Nigeria across all government-owned investments due to funding challenges, corruption, and unavailability of adequate maintenance framework. To ensure the reliability of power system infrastructure from generation to distribution, effective maintenance strategy encompassing equipment maintenance timeline and compliance tracking system must be developed. The procurement and installation of sub-standard equipment must be prevented at all levels as there may be tendencies for sabotage and compromise during tendering and bidding processes.

Setting up a state-of-the-art power system requires adequate funding, and Nigerian investors are either unable or unwilling to invest significantly in the power sector. The Federal Government through relevant ministries must develop policies for securing external funding with reasonable and mutually beneficial terms. Investment disputes and contract award failures like the case between the Process and Industrial Developments Limited and the Federal Government of Nigeria on the conversion of wet gas to lean gas for powering the electric grid must be prevented in the future because this discourages other potential investors in the power sector.

Continuity in policy is vital for seeing plans to completion. Change of government in Nigeria is often followed by change of policies and goals, and this usually results in abandoned projects resulting in wasted national funds. A team of experts in relevant fields need to be setup to develop a national power sector development strategy that covers the maintenance of existing facility and expansion projects. The capacity of the power transmission network in Nigeria is estimated at 7,500MW with transmitted power usually below 5,000MW, and as such, even if generation goes above this capacity, the transmission network cannot handle it, and a similar challenge exists on the distribution network. Hence, a successful power sector development plan must consider the challenges at each stage of the power system and develop strategies for mitigating the identified issues

The design of the power transmission and distribution network in Nigeria is based on decades-old approach and may not meet the requirements of this modern time. The routing, the length and sometimes the cable type can support increased line losses resulting in wastage of insufficient electrical energy, and this must be corrected using modern techniques like the compact line technology. A number of transformers, circuit breakers etc. were installed several decades ago and are no more optimal but unfortunately, such equipment are even being overloaded, and this often results in system failure and power instability. These issues must be addressed appropriately and funding must be put in place to replace such old equipment in milestones until all are phased-out. To reduce dependency on the national grid, there is a need to promote the use of renewable energy sources and to setup arrangement with banks that will facilitate long term loans toward motivating interested households to setup alternative energy sources in their homes.

Energy theft, unmetered customers, and ATC&C losses generally, are sources of income losses in the power sector. Customer education and orientation must be deployed as a strategy through a rigorous campaign. Resolving the multitude of challenges facing the power sector in Nigeria which include insufficient funds, sabotage, gas pipeline vandalization, energy theft, energy wastage, unmetered customers etc. will require more than words. The commitment of all stakeholders to finding a sustainable solution is the first step towards achieving this objective.

VI. Conclusion

In this study, we evaluated the performance of the Nigerian power sector in terms of the power generation and the Aggregate Technical, Commercial and Collection Losses using historical data. The result of the analysis revealed that from 2015 to 2018 power generation has not significantly improved. Also, both economic and technical losses are still very high and far above the 20% target. Power generation from 2016 to 2018 shows very sharp fluctuations which is an indication of generation instability due to the usual gas supply disruptions, equipment failures and long duration of maintenance activities requiring shutdown without adequate standby turbines or generators. Nigerians are tired of epileptic power supply and exorbitant charges for unused energy due to estimated billing by DisCos as a result of their inability to provide meters for all customers as statutorily required. The way forward is currently not clear because the various efforts of the government has not yielded the desired turnaround.

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